Efficient Diffusion Language Models: A Comprehensive Survey

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GitHub Repository: https://github.com/MessiX77/Awesome-Efficient-dLLMs

Abstract

Diffusion language models (dLLMs) have recently emerged as a promising alternative to autoregressive (AR) language models, offering competitive performance with a fundamentally different generation paradigm. Instead of producing tokens strictly left-to-right, dLLMs iteratively refine partially masked sequences, enabling bidirectional context modeling and supporting parallel token updates. This design creates new opportunities for accelerating generation and improving controllability. However, practical deployment of dLLMs remains challenging due to the high cost of training at scale and the lack of mature inference optimizations such as cache-friendly decoding and robust parallel sampling. In this work, we provide a comprehensive overview of recent progress on *efficient* dLLMs. To reflect how efficiency bottlenecks arise across the full lifecycle of dLLMs, spanning training, decoding, and deployment, we organize existing approaches into five categories: *Training*, *Inference*, *Context*, *Framework*, and *Multimodality*. For each category, we summarize representative methods and highlight their motivations, key techniques, and empirical trade-offs. Finally, we discuss open challenges and outline promising directions for future research toward high-throughput, robust, and practically deployable diffusion language models.

1 Introduction

Autoregressive (AR) large language models (LLMs) have shown strong performance across tasks, but their left-to-right decoding generates tokens sequentially, limiting parallelism and making inference inefficient. Diffusion language models (dLLMs) offer an alternative paradigm by iteratively refining partially masked sequences. Early text diffusion models such as DiffuSeq (Gong et al., 2023), SSD-LM (Han et al., 2023), and AR-Diffusion (Wu et al., 2023) operate in a continuous or latent space, whereas the models since scaled to billions of parameters are predominantly discrete, corrupting text by masking tokens and recovering them by iterative unmasking and reaching competitive quality with strong AR counterparts. This distinction matters for efficiency, since masking defines the unit of parallelism and determines what can be cached across denoising steps. We therefore use *dLLM* for these modern discrete models, and mark continuous formulations explicitly where they appear. This paradigm enables bidirectional context modeling and allows multiple tokens to be updated in parallel, making dLLMs a promising direction for efficient generation and improving controllability (Yang et al., 2025a; You et al., 2025; Zhu et al., 2025a; Liu et al., 2025a).

However, deploying dLLMs in practice still faces several serious efficiency bottlenecks. First, training dLLMs from scratch is highly data- and compute-intensive, and early open-source models (e.g., LLaDA (Nie et al., 2025)) remain less competitive than strong AR counterparts. Second, during inference, dLLMs have not yet shown consistent speed advantages over AR models with optimized serving engines like vLLM. This is largely due to the incompatibility of standard KV caching with bidirectional attention and the generation quality degradation of parallel decoding (Wu et al., 2026b). Third, supporting long-context and variable-length generation is nontrivial, as many dLLMs rely on fixed denoising schedules and pre-specified output lengths (Wu et al., 2026a).

In this survey, we provide the first review organized around the efficiency of dLLMs and their practical deployment. As illustrated in Figure 1, we organize the literature along the full lifecycle of a dLLM, where each stage exposes a distinct efficiency bottleneck: training cost (*Training*), decoding latency (*Inference*), sequence-length budget (*Context*), system-level throughput (*Framework*), and cross-modal scaling (*Multimodality*). Con-

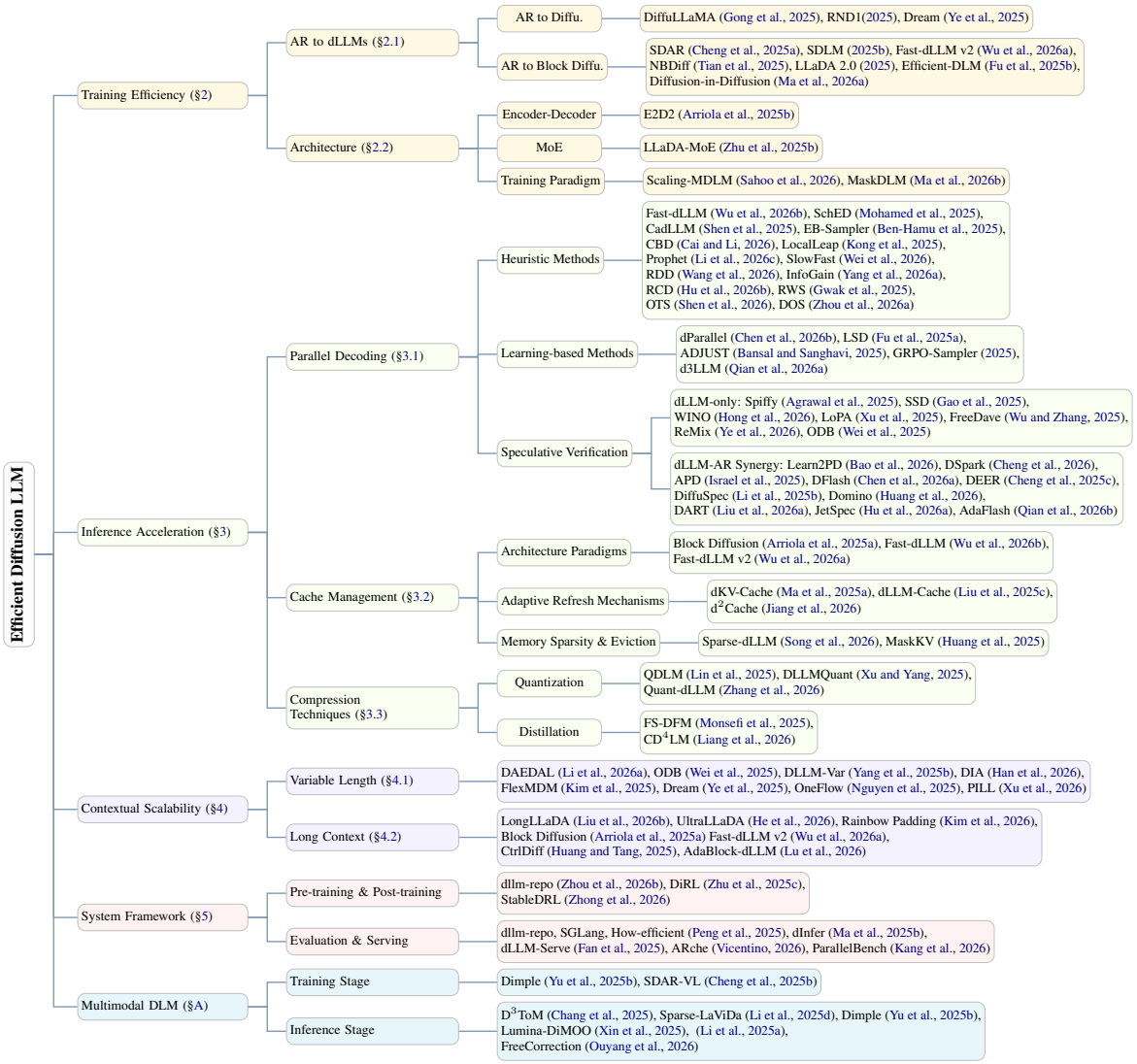


Figure 1: Our taxonomy tree of Efficient Diffusion Language Models (dLLMs).

cretely, for *training efficiency*, we summarize approaches that convert pretrained AR models into dLLMs (Ye et al., 2025; Wu et al., 2026a) as well as architectural designs (Zhu et al., 2025b) that enable efficient dLLMs. For *inference efficiency*, we review parallel decoding strategies (Wu et al., 2026b; Chen et al., 2026b), dynamic KV-cache management (Ma et al., 2025a; Jiang et al., 2026), and compression techniques (Lin et al., 2025; Zhang et al., 2026) that improve the speed-quality trade-off. We further discuss *context scalability*, including methods for variable-length and long-context generation (Liu et al., 2026b; Arriola et al., 2025a). In addition, we highlight emerging *frameworks* that support dLLM training, evaluation, and production serving (Zhou et al., 2026b; Ma et al., 2025b), and summarize recent progress on efficient *multimodal diffusion language models* (Yu et al., 2025b).

Through this lens, three high-level challenges recur across categories and motivate our taxonomy: (i) the mismatch between bidirectional refinement and AR-friendly system primitives such as KV cache and prefix prefill; (ii) the throughput and quality trade-off under aggressive parallel decoding, where parallelism gains are easily offset by lost coherence; and (iii) the lack of mature open-source infrastructure that ports gains from algorithm papers to production-level latency.

Although existing surveys (Li et al., 2025e; Yu et al., 2025a) have reviewed diffusion language models broadly, our survey focuses specifically on efficiency-related research. A detailed comparison is provided in Appendix Table 2.

Our main contributions are threefold: (1) we propose a unified taxonomy that organizes efficient dLLM techniques into five categories, including

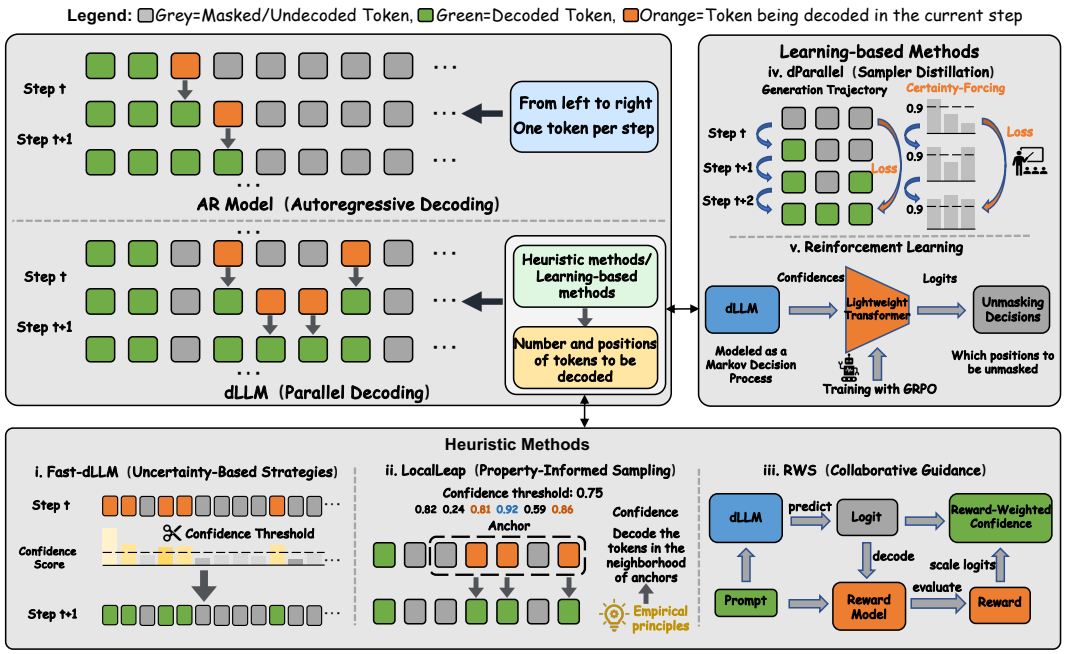


Figure 2: **Illustration of Parallel Decoding for dLLMs.** Unlike AR models that decode one token per step in a left-to-right manner, dLLMs can update multiple tokens simultaneously. In practice, heuristic or learning-based strategies are used to decide *which* token positions to decode and *how many* tokens to update at each step.

Training, *Inference*, *Context*, *Framework*, and *Multimodality*; (2) we systematically review representative works in each category and analyze how they address key bottlenecks; and (3) we outline open challenges and promising future directions for efficient dLLMs.

2 Training Efficiency

2.1 AR to dLLMs

AR to Diffusion. Training diffusion language models from scratch at scale is computationally expensive. A practical alternative is to start from a pretrained AR model and focus on the core challenge of transition: how to convert an AR model into a diffusion-ready backbone without losing its linguistic competence. DiffuLLaMA (Gong et al., 2025) explicitly targets this AR-to-diffusion transfer, introducing dedicated transition techniques, such as attention-mask annealing, shift operations, and a time-embedding-free architecture. Dream-7B (Ye et al., 2025) pushes this direction further by turning the transition into a full training pipeline, notably incorporating context-adaptive token-level noise scheduling to improve downstream performance. In contrast, RND1 (Chandrasegaran et al., 2025) argues that the transition can be simplified: it directly switches from AR initialization to dLLM training, avoiding explicit intermediate adaptation steps (e.g., attention annealing).

AR to Block Diffusion. Recent block diffusion language models (Arriola et al., 2025a) generate text block-by-block while denoising tokens within each block, preserving a structured dependency pattern aligned with AR pretraining. Adapting AR models to block-wise diffusion is therefore substantially more efficient than fully diffusion conversion. SDAR (Cheng et al., 2025a) is the first to propose an efficient AR-to-block diffusion conversion pipeline, leveraging dense token-level supervision inherited from AR pretraining so that the diffusion objective aligns within a short adaptation stage. SDLM (Liu et al., 2025b) introduces a Next Sequence Prediction (NSP) objective that interleaves noise blocks with target blocks, enabling parallel training while retaining block structure. NBDiff (Tian et al., 2025) designs a context-causal attention mask tailored for block diffusion adaptation and adds an auxiliary AR loss for convergence and knowledge retention. Fast-dLLM v2 (Wu et al., 2026a) and Efficient-DLM (Fu et al., 2025b) show that effective block diffusion models can be obtained through lightweight post-training or continuous pretraining initialized from AR weights. LLaDA 2.0 (Bie et al., 2025) stabilizes the transition with a Warmup-Stable-Decay schedule, and Diffusion-in-Diffusion (Ma et al., 2026a) restores global coherence by refining small-block drafts with a larger bidirectional receptive field.

2.2 Architecture

Architecture choices in dLLMs are driven by efficiency: diffusion training and inference repeatedly process partially noised sequences, so designs either (i) decouple clean and noisy representations to avoid redundant compute, or (ii) scale capacity through sparsity. E2D2 (Arriola et al., 2025b) adopts an **encoder-decoder** layout for block diffusion, where the encoder processes clean inputs and the decoder denoises, cutting training cost by roughly half versus equal-size decoder-only models. Orthogonally, **Mixture-of-Experts** (MoE) routes computation through sparse experts: LLaDA-MoE (Zhu et al., 2025b) is the first MoE diffusion language model, outperforming dense diffusion baselines and matching AR models while activating far fewer parameters at inference. Beyond architecture, recent work also revisits training: (Sahoo et al., 2026) compares the scaling behavior and speed-quality trade-offs of masked, uniform-state, and interpolating diffusion models, while MaskDLM (Ma et al., 2026b) introduces information-density-aware masking to improve training effectiveness.

3 Inference Acceleration

3.1 Parallel Decoding

Open-source dLLMs have yet to fully realize their high-throughput potential, primarily because decoding multiple tokens in parallel can degrade generation quality. Existing strategies broadly fall into three families: heuristic rules, learned samplers, and speculative verification.

Parallel decoders can be compared along three common design choices (Fig. 2): *which* masked positions to commit, *how many* tokens to commit at each step, and *whether* previous commitments can be revised. Fast-dLLM (Wu et al., 2026b), for example, uses a fixed confidence threshold to select tokens for irreversible commitment. Its effectiveness therefore depends on confidence calibration, while over-confident early predictions may propagate errors to later steps. From this perspective, existing methods modify different parts of the decoding process: schedule-based methods control how many tokens are committed, property-informed methods refine which positions are selected, learned samplers replace hand-crafted selection rules, and revisable or verification-based methods relax the assumption of irreversible commitment.

Heuristic Methods leverage handcrafted metrics and empirical observations to optimize decoding, offering plug-and-play gains without retraining. We group them into three sub-strategies:

- **Uncertainty-Based Strategies**: These methods use token-level uncertainty signals (e.g., confidence or entropy) to decide *which* tokens to decode and *how many* to update per step. Fast-dLLM (Wu et al., 2026b) unmasks tokens whose confidence exceeds a preset threshold; SchED (Mohamed et al., 2025), CadLLM (Shen et al., 2025), and EB-Sampler (Ben-Hamu et al., 2025) extend this idea with progress-aware schedules, dynamic block/step sizes, and entropy-bounded budgets, respectively. CBD (Cai and Li, 2026) further demonstrates that confidence-based decoding is sample-efficient under mild assumptions.
- **Property-Informed Sampling**: Rather than relying on uncertainty scores, these methods exploit empirical decoding patterns such as locality, progressive consistency, and early stabilization. LocalLeap (Kong et al., 2025) parallelizes within neighborhoods around high-confidence anchor tokens; Prophet (Li et al., 2026c) terminates once token transitions stabilize; Slow-Fast Sampling (Wei et al., 2026) cycles between exploration and acceleration phases guided by certainty, convergence, and positional principles. RDD (Wang et al., 2026) further makes decoding revocable so that wrongly committed tokens can be rolled back. InfoGain (Yang et al., 2026a) chooses positions by information gain rather than confidence. RCD (Hu et al., 2026b) recycles low-confidence tokens dropped by remasking as contextual residuals for the next denoising step.
- **Collaborative Guidance**: these methods keep a single decoding pass but let an external signal reshape the joint distribution. RWS (Gwak et al., 2025) uses a reward model to score candidate updates. OTS (Shen et al., 2026) jointly searches over generation order and token choices. DOS (Zhou et al., 2026a) exploits token dependencies to accelerate information propagation.

Learning-based Methods cover sampler distillation, trained auxiliary modules, and reinforcement learning, yielding larger speedups in both semi-AR and fully parallel settings:

- **Sampler Distillation**: dParallel (Chen et al., 2026b) introduces certainty-forcing distillation, training the model to reach high certainty over

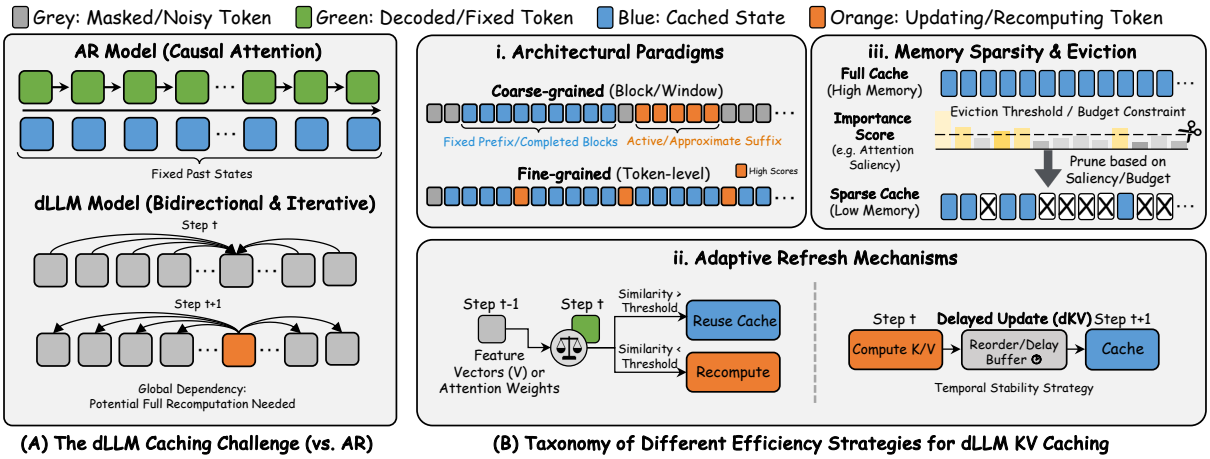


Figure 3: **Illustration of Cache Management for dLLMs.** (A) The key challenge in dLLM caching. Unlike AR models, dLLMs rely on bidirectional attention and iteratively refine the entire sequence, so KV states are not fixed and cannot be reused. (B) A taxonomy of efficiency strategies for dLLM KV caching: (i) *granularity control*, from coarse-grained blocks to fine-grained tokens; (ii) *adaptive refresh*, using similarity thresholds or delayed updates; and (iii) *memory sparsity and eviction*, guided by saliency signals and budget constraints.

masked tokens earlier and in parallel. LSD (Fu et al., 2025a) aligns a few-step student sampler with the score trajectory of a many-step teacher, and LSD+ further learns a non-uniform schedule. d3LLM (Qian et al., 2026a) pushes this further via pseudo-trajectory distillation for ultra-low steps.

- **Trained Auxiliary Modules**: ADJUST (Bansal and Sanghavi, 2025) trains a lightweight single-layer sampler atop a frozen dLLM; multiple forward passes of this sampler after one full pass approximately sample multiple tokens from the joint distribution.
- **Reinforcement Learning**: Jazbec et al. (2025) cast dLLM sampling as a Markov Decision Process and use GRPO to train a lightweight policy network that adaptively unmasks token positions per step.

Speculative Verification separates proposal from validation: multiple token updates are drafted in parallel and then checked by a verification rule, with only validated proposals committed to the sequence. Unlike the methods above, which directly determine which predictions to commit, speculative methods allow draft proposals to be rejected during verification, providing an additional safeguard against unreliable parallel updates.

- **dLLM-only Speculation** drafts and verifies within one dLLM. Spiffy (Agrawal et al., 2025) predicts future intermediate states during denoising and organizes them into "Directed Draft Graphs" calibrated offline for higher acceptance rates. SSD (Gao et al., 2025) self-drafts multiple tokens and verifies them with a hierarchical tree, allowing valid sub-sequences to be accepted in one forward pass. WINO (Hong et al., 2026), LoPA (Xu et al., 2025), and FreeDave (Wu and Zhang, 2025) generate multiple candidates and prune them using bidirectional context or self-verification. ReMix (Ye et al., 2026) instead validates drafts through rejection mixing. ODB-dLLM (Wei et al., 2025) reduces the cost of each drafting round through step skipping and cross-layer KV-cache sharing.

- **dLLM-AR Synergy** combines the diffusion and autoregressive paradigms, with one model drafting and the other assisting verification. APD (Israel et al., 2025) uses a small AR model to score the joint plausibility of dLLM drafts and adapt the parallel lookahead. Learn2PD (Bao et al., 2026) trains a lightweight filter to predict draft correctness and re-mask low-confidence tokens for selective re-denoising. DEER (Cheng et al., 2025c) uses a dLLM as a block drafter and an AR target for verification, reducing uncertainty accumulation. DiffuSpec (Li et al., 2025b) extracts AR-aligned sequences from the dLLM token lattice through training-free Causal-Consistency Path Search. DFlash (Chen et al., 2026a) conditions a lightweight diffusion drafter on hidden features from a large AR target. DART (Liu et al., 2026a) instead brings diffusion-style drafting to AR inference.

Discussion. Heuristic methods are simple and plug-

and-play, but their effectiveness depends on the reliability of uncertainty signals. Learning-based methods provide a more systematic route to faster decoding through direct policy optimization, at the cost of additional training and tuning. Speculative verification reduces the reliance on calibrated unmasking signals by validating drafts before commitment. This provides a safeguard against incorrect parallel updates, but introduces additional computation for verification.

3.2 Cache Management

Standard Key-Value (KV) caching is incompatible with the bidirectional attention used by dLLMs: token representations are refined across denoising steps, so cached states cannot be naively reused. As shown in Figure 3, current works tackle this through architectural adaptations, algorithmic refinements, and memory optimizations.

Block Diffusion (Arriola et al., 2025a) enables exact KV reuse across blocks, but does not eliminate the caching problem within each active block. Tokens in the active block are still refined bidirectionally and must be recomputed, while the block size is fixed before deployment rather than adapted to the input or denoising state. Subsequent methods relax these constraints along two main axes: the *granularity* at which states can be reused within a block, and the *staleness* tolerated before cached states must be refreshed.

Architecture Paradigms. Architectural approaches differ in the granularity at which generation dependencies are fixed and reused (Fig. 3(i)). Coarse-grained methods such as Block Diffusion (Arriola et al., 2025a) adopt a window-based scheme that caches completed prefixes as fixed "Blue" blocks while restricting diffusion to an active "Orange" suffix. Fast-dLLM (Wu et al., 2026b) and Fast-dLLM v2 (Wu et al., 2026a) refine this with a DualCache design that separates exact prefix states from approximate masked-suffix representations. Fine-grained strategies operate at the token level, selectively recomputing only a few unstable tokens identified by confidence or attention while reusing cached states for stable ones.

Adaptive Refresh Mechanisms. For standard diffusion architectures, the main caching challenge is KV drift across denoising steps (Fig. 3(ii)). These methods dynamically decide whether cached states should be reused or recomputed based on stability signals such as feature similarity or attention. dLLM-Cache (Liu et al., 2025c) compares Value vectors against a similarity threshold (V-Verify), reusing the cache when changes are small. dKV-Cache (Ma et al., 2025a) delays cache commits until token states stabilize. d^2Cache (Jiang et al., 2026) prioritizes recomputation for high-attention groups while freezing low-attention states.

Memory Sparsity & Eviction. For long-context generation, caches must operate under strict memory budgets (Fig. 3(iii)). These methods (Nguyen-Tri et al., 2026) retain only the most informative tokens. Sparse-dLLM (Song et al., 2026) assigns attention-based importance scores and dynamically evicts low-importance entries; MaskKV (Huang et al., 2025) extends this with a *Mask as Prophet* principle, aggregating importance signals from mask tokens to guide eviction.

Discussion. The three groups address different aspects of cache reuse. Architectural methods determine the granularity of reuse, adaptive-refresh methods decide when cached states have become too stale, and eviction methods determine which states to retain under a memory budget. Despite these differences, they face a common trade-off between more aggressive reuse and the errors introduced by stale states. Current methods typically control this trade-off using empirically tuned, model-specific criteria. A more general direction is therefore to develop reuse or refresh criteria that transfer across model architectures, sequence lengths, and denoising steps.

3.3 Compression Techniques

Quantization. To mitigate memory and computational overhead, Post-Training Quantization (PTQ) (Yao et al., 2026; Yang et al., 2026b; Lin et al., 2026) has been extended from AR LLMs to dLLMs. QDLM (Lin et al., 2025) establishes the first comprehensive benchmark, identifying distinct activation outliers in dLLMs and evaluating SOTA PTQ baselines. Subsequent works address dLLM features: DLLMQuant (Xu and Yang, 2025) introduces fine-grained quantization that accounts for temporal embedding, mask ratios, and attention distributions; Quant-dLLM (Zhang et al., 2026) identifies the role of timestep-dependent masking and proposes a data-aware, any-order quantizer that supports extreme 2-bit weight-only settings. Two practical gaps remain: (1) the lack of low-bit kernels that turn theoretical compression into wall-

clock speedups, and (2) the unaddressed synergy between quantization and dLLM-specific caching.

Distillation. To bridge the sampling-speed gap with AR LLMs, several works compress the iterative denoising process into a few steps. FS-DFM (Monsefi et al., 2025) uses few-step discrete flow-matching to attain competitive long-text perplexity in eight steps, and CD^{4}LM (Liang et al., 2026) applies discrete-space consistency distillation so that the model denoises correctly even when intermediate steps are skipped. Preserving long-form coherence and reasoning under aggressive step reduction remains the central challenge, and joint exploration with pruning is still underexplored.

Discussion. Quantization and distillation target different costs: quantization reduces memory and bandwidth, whereas distillation reduces the number of denoising steps. The two are largely orthogonal: a low-bit backbone still follows the full denoising schedule, while a few-step student may still use full-precision weights. Whether they can be combined without compounding errors remains to be explored, as both low-bit logits and aggressive step reduction can affect the confidence signals used by parallel decoders.

> ❖ **Takeaway** (Appendix Table 1):
> • Learning-based decoding reports the largest speedups.
> • Training-free cache reports composable 2–10× gains.
> • In practice, combining these axes is likely necessary.

4 Contextual Scalability

Sequence length matters here as an efficiency cost rather than a capability. Most dLLMs denoise a fixed-length canvas, so the length budget directly affects computation: short outputs may still incur computation on masked padding positions, while longer sequences increase the cost of bidirectional attention and the activation memory required during denoising. The following two subsections address these costs in turn.

4.1 Variable Length

Early dLLMs rely on a fixed, pre-specified output length, which wastes computation on padding tokens. Native variable-length generation lets compute adapt to the actual response length while preserving diffusion's parallel decoding advantages.

Inference-Time Adaptive Length Control. This line of work determines output length on the fly during inference using confidence-based stopping signals, rather than fixing it in advance. DAEDAL (Li et al., 2026a) starts from a short sequence, adjusts length based on End-Of-Sequence (EOS) token confidence, and then inserts masks at low-confidence positions for iterative refinement. dLLM-Var (Yang et al., 2025b) enforces explicit EOS modeling and conditionally appends new blocks when no EOS is produced, enabling open-ended generation. PILL (Xu et al., 2026) predicts infilling lengths via lightweight probing and avoids costly iterative length search for efficient adaptive-length decoding.

Dynamic-Canvas and Any-Order Generation. This paradigm frames generation as iterative refinement on a dynamic canvas, where length emerges implicitly from insertion and editing rather than explicit length control. FlexMDM (Kim et al., 2025) decouples length from token prediction by first estimating insertion counts between existing positions and then filling the inserted masks via posterior denoising, enabling flexible-length generation under non-causal diffusion. Dream (Ye et al., 2025) combines autoregressive initialization with diffusion-based infilling to support the generation and completion of text segments with arbitrary lengths. OneFlow (Nguyen et al., 2025) generalizes this to multimodal via an edit-flow mechanism for parallel, variable-length synthesis.

Discussion. These approaches make variable-length generation practical for efficient dLLMs. The next step is more robust length-adaptive decoding that maintains stable termination and coherence under aggressive parallelism.

4.2 Long Context

Despite significant progress in AR LLM for long-context processing, dLLMs have historically struggled with context scalability due to fundamental architectural mismatches. Recent advances have enabled context windows to expand from thousands to over 128K tokens while preserving generation quality, primarily through following techniques:

- **Position Encoding and Extrapolation:** LongLLaDA (Liu et al., 2026b) discovers that dLLMs degrade more gracefully beyond their training context and retain strong sensitivity to recent context segments. Based on this, it derives a training-free RoPE scaling rule by computing the critical dimension and scaling factor, enabling

effective context extension without parameter modification. UltraLLaDA (He et al., 2026) further extends the context window to 128K tokens through diffusion-aware NTK position encoding and a multi-document concatenation masking strategy during post-training.

- **Blockwise Diffusion:** Concurrent methods decompose extended sequences into segments and perform parallel denoising within blocks while coordinating cross-block dependencies. Block Diffusion (Arriola et al., 2025a) combines sequential block generation with intra-block parallel refinement to maintain coherence at scale; Fast-dLLM v2 (Wu et al., 2026a) further improves long-context efficiency through bidirectional block modeling and hierarchical caching. CtrlDiff (Huang and Tang, 2025) and AdaBlock-dLLM (Lu et al., 2026) adapt block granularity based on semantic or attention cues, allocating compute where long-range dependencies are most critical. Complementarily, Rainbow Padding (Kim et al., 2026) mitigates early termination in instruction-tuned dLLMs via cyclic multi-token padding that prevents premature EOS prediction.

Discussion. These advances establish a foundation for long-context dLLMs, but realizing their full benefit needs deeper co-design across acceleration techniques to keep global coherence under extreme context lengths.

5 System Framework

As dLLMs move toward large-scale deployment, the focus has shifted from algorithmic feasibility to system-level efficiency. While vLLM and SGLang serve as industry standards for AR LLMs, the dLLM ecosystem is still nascent. Current progress integrates dLLM support into existing engines and develops dedicated frameworks:

- **Unified Training and Evaluation**: dllm-repo (Zhou et al., 2026b) standardizes the full diffusion LLM lifecycle, including training, fine-tuning, and evaluation. ARche (Vicentino, 2026) further provides a controlled comparison between AR and masked diffusion models, isolating the actual benefit of diffusion under matched compute. ParallelBench (Kang et al., 2026) complements these by stress-testing the throughput-quality trade-off of parallel decoding.
- **Post-training Optimization**: DiRL (Zhu et al., 2025c) addresses the training-inference mismatch in dLLMs via online policy updates and training-inference co-design. StableDRL (Zhong et al., 2026) complements this by stabilizing RL for dLLMs, mitigating the variance and policy-collapse issues observed in dLLM RL.
- **Mainstream Serving Integration**: SGLang has introduced initial support for the LLaDA 2.0 series, leveraging its Chunked-Prefill pipeline to support 16B and 100B Block Diffusion LLMs.
- **Specialized Inference Engines**: dInfer (Ma et al., 2025b) develops a modular pipeline that decouples model logic from diffusion iteration management, achieving a 10× speedup over baselines via CUDA Graphs and loop unrolling.
- **Production Serving**: dLLM-Serve (Fan et al., 2025) addresses the highly variable memory footprint of inference via Logit-Aware Activation Budgeting and a Phase-Multiplexed Scheduler.

Discussion. These frameworks form a growing ecosystem covering the full dLLM deployment lifecycle, from single-request latency to high-concurrency serving. As dLLMs move toward more complex non-AR architectures and multimodal settings, the community still needs production-grade open-source frameworks with portable kernel support and seamless integration with mainstream serving infrastructure.

6 Future Research Directions

We close with concrete open problems, distilled from recurring gaps surfaced across Sections 2–5. Some gaps emerge only at the intersection of categories. Techniques developed for different stages may act on the same signals or system constraints, creating interactions that are missed when they are optimized in isolation. For example, quantization can alter the confidence scores used by parallel decoders, learned samplers may depend on decoding schedules that vary across serving settings, and cache policies may waste computation or memory on padded or low-utility positions. Appendix Table 3 summarizes these interactions and the design questions they expose. Such interactions remain underexplored because most existing methods and evaluations are developed within individual efficiency categories.

Training Efficiency. Several questions stand out. First, how to scale AR-to-block diffusion conversion to 100B+ parameters with stable optimization,

given that current recipes still need a long warmup stage (Bie et al., 2025; Fu et al., 2025b). Second, whether a clean scaling law exists for masked diffusion akin to Chinchilla for AR (Ye et al., 2025); without one, allocating data, parameters, and noise schedules remains an empirical exercise. A third concrete direction is curriculum design that exploits bidirectional conditioning: synthesizing pretraining data with strong long-range structure (e.g., long-document QA, code with cross-file dependencies) so that bidirectional attention pays off at training time rather than only at inference.

Inference Optimization. Three open problems are relatively well-defined: (i) a unified throughput-quality knob for parallel decoding, since the current confidence/entropy/local-determinism heuristics are model- and prompt-specific (Wu et al., 2026b; Kong et al., 2025); (ii) a diffusion-native KV cache primitive that handles bidirectional attention with the same composability as PagedAttention or FlashAttention (Kwon et al., 2023); (iii) low-bit kernels for non-uniform activation patterns observed in dLLMs (Lin et al., 2025), including INT4/INT2 GEMM kernels that fuse step-skipping with quantization. Closing any one of these would let a single dLLM checkpoint match an AR engine in wall-clock latency, which no current paper has demonstrated end-to-end.

Context and Framework. The gap to AR systems with 1M-token windows is concrete: dLLMs need a memory layout in which bidirectional attention is sparsified per block while preserving global coherence. Blockwise diffusion with adaptive block sizes is a natural candidate, but currently lacks an analogue of FlashAttention-2 to make the kernel competitive at 128K+. On the system side, the ecosystem still lacks a single open-source stack that covers pretraining, RL post-training, and serving in one repository with reproducible numbers, making cross-paper comparison difficult.

Multimodal DLM. Discrete diffusion provides a unified token space across modalities, which makes two directions tractable. First, token-level rewards become natural under discrete masks, so RL post-training can target spatial-semantic alignment (e.g., precise visual grounding, document parsing) at fine granularity (Yang et al., 2025a; Xin et al., 2025; Liu et al., 2026c). In addition, the efficiency of multimodal diffusion language models could benefit from modality-aware decoding that adapts step counts, block sizes, and cache budgets to the different redundancy patterns of text and visual tokens, ideally co-designed with hardware kernels.

Beyond Masked Transformers. Two assumptions behind most current work may not hold for long. The first is the transformer backbone: linear attention, sub-linear token mixers, and hybrid recurrent-attention stacks change the cost of bidirectional refinement, so it is an open question whether the cache and parallel-decoding techniques surveyed here still pay off once attention is no longer quadratic. The second is discrete masking. Continuous formulations remain viable, and DLM-One (Chen et al., 2025) shows that distilling a score function into a one-step generator in embedding space can remove the iterative loop entirely. Whether such generators can match the controllability and infilling behaviour of mask-based decoding is unresolved and a positive answer would turn step-count reduction into a question of formulation rather than scheduling.

7 Conclusion

This survey provides a comprehensive overview of existing research efforts toward improving the efficiency of dLLMs. To capture efficiency bottlenecks across the dLLM pipeline, from pretraining and decoding to long-context scalability and multimodal extension, we organize the literature into five categories, summarize representative methods, and discuss their practical implications. Finally, we highlight open challenges and future directions, with the hope of inspiring further progress toward efficient and widely deployable dLLMs.

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Limitations

This survey focuses specifically on the efficiency of diffusion language models, and a few scope choices are worth noting. Due to the page limit, our taxonomy highlights representative methods rather than exhaustively listing every related work; some closely related techniques are placed in the

appendix or in the taxonomy figure. We also summarize empirical trade-offs as reported in the original papers rather than performing a unified re-evaluation, since existing works use heterogeneous backbones, hardware, and metrics; a controlled benchmark across the full taxonomy is left to future work. Finally, our discussion centers on the efficiency of text and multimodal dLLMs, and does not deeply cover orthogonal topics such as alignment and safety, which deserve dedicated studies.

References

Sudhanshu Agrawal, Risheek Garrepalli, Raghavv Goel, Christopher Lott, Fatih Porikli, and Mingu Lee. 2025. Structuring the future: Diffusion llm speculative decoding via calibrated draft graphs. *arXiv preprint arXiv:2509.18085*.

Marianne Arriola, Aaron Gokaslan, Justin Chiu, Zhihan Yang, Zhixuan Qi, Jiaqi Han, Subham Sahoo, and Volodymyr Kuleshov. 2025a. Block diffusion: Interpolating between autoregressive and diffusion language models. In *International Conference on Learning Representations*, volume 2025, pages 50726–50753.

Marianne Arriola, Yair Schiff, Hao Phung, Aaron Gokaslan, and Volodymyr Kuleshov. 2025b. Encoder-decoder diffusion language models for efficient training and inference. In *Advances in Neural Information Processing Systems*, volume 38, pages 15612–15644. Curran Associates, Inc.

Parikshit Bansal and Sujay Sanghavi. 2025. Enabling approximate joint sampling in diffusion lms. *arXiv preprint arXiv:2509.22738*.

Wenrui Bao, Zhiben Chen, Dan Xu, and Yuzhang Shang. 2026. Learning to parallel: Accelerating diffusion large language models via learnable parallel decoding. In *The Fourteenth International Conference on Learning Representations*.

Heli Ben-Hamu, Itai Gat, Daniel Severo, Niklas S Nolte, and Brian Karrer. 2025. Accelerated sampling from masked diffusion models via entropy bounded unmasking. *Advances in Neural Information Processing Systems*, 38:55981–56007.

Tiwei Bie, Maosong Cao, Kun Chen, Lun Du, Mingliang Gong, Zhuochen Gong, Yanmei Gu, Jiaqi Hu, Zenan Huang, Zhenzhong Lan, et al. 2025. Llada2. 0: Scaling up diffusion language models to 100b. *arXiv preprint arXiv:2512.15745*.

Changxiao Cai and Gen Li. 2026. Confidence-based decoding is provably efficient for diffusion language models. *arXiv preprint arXiv:2603.22248*.

Keshigeyan Chandrasegaran, Armin W. Thomas, Jerome Ku, Federico Berto, Jae Myung Kim, Garyk Brixi, Eric Nguyen, Stefano Massaroli, and Michael Poli. 2025. Rnd1: Simple, scalable ar-to-diffusion conversion.

Shuochen Chang, Xiaofeng Zhang, Qingyang Liu, and Li Niu. 2025. D^3tom: Decider-guided dynamic token merging for accelerating diffusion mllms. *arXiv preprint arXiv:2511.12280*.

Jian Chen, Yesheng Liang, and Zhijian Liu. 2026a. Dflash: Block diffusion for flash speculative decoding. *arXiv preprint arXiv:2602.06036*.

Tianqi Chen, Shujian Zhang, and Mingyuan Zhou. 2025. Dlm-one: Diffusion language models for one-step sequence generation. *arXiv preprint arXiv:2506.00290*.

Zigeng Chen, Gongfan Fang, Xinyin Ma, Ruonan Yu, and Xinchao Wang. 2026b. dparallel: Learnable parallel decoding for dLLMs. In *The Fourteenth International Conference on Learning Representations*.

Shuang Cheng, Yihan Bian, Dawei Liu, Linfeng Zhang, Qian Yao, Zhongbo Tian, Wenhai Wang, Qipeng Guo, Kai Chen, Biqing Qi, et al. 2025a. Sdar: A synergistic diffusion-autoregression paradigm for scalable sequence generation. *arXiv preprint arXiv:2510.06303*.

Shuang Cheng, Yuhua Jiang, Zineng Zhou, Dawei Liu, Wang Tao, Linfeng Zhang, Biqing Qi, and Bowen Zhou. 2025b. Sdar-vl: Stable and efficient block-wise diffusion for vision-language understanding. *arXiv preprint arXiv:2512.14068*.

Xin Cheng, Xingkai Yu, Chenze Shao, Jiashi Li, Yunfan Xiong, Yi Qian, Jiaqi Zhu, Shirong Ma, Xiaokang Zhang, Jiasheng Ye, et al. 2026. Dspark: Confidence-scheduled speculative decoding with semi-autoregressive generation. *arXiv preprint arXiv:2607.05147*.

Zicong Cheng, Guo-Wei Yang, Jia Li, Zhijie Deng, Meng-Hao Guo, and Shi-Min Hu. 2025c. Deer: Draft with diffusion, verify with autoregressive models. *arXiv preprint arXiv:2512.15176*.

Jiakun Fan, Yanglin Zhang, Xiangchen Li, and Dimitrios S Nikolopoulos. 2025. Taming the memory footprint crisis: System design for production diffusion llm serving. *arXiv preprint arXiv:2512.17077*.

Feiyang Fu, Tongxian Guo, and Zhaoqiang Liu. 2025a. Learnable sampler distillation for discrete diffusion models. In *Advances in Neural Information Processing Systems*, volume 38, pages 70161–70190. Curran Associates, Inc.

Yonggan Fu, Lexington Whalen, Zhifan Ye, Xin Dong, Shizhe Diao, Jingyu Liu, Chengyue Wu, Hao Zhang, Enze Xie, Song Han, et al. 2025b. Efficient-dlm: From autoregressive to diffusion language models, and beyond in speed. *arXiv preprint arXiv:2512.14067*.

Yifeng Gao, Ziang Ji, Yuxuan Wang, Biqing Qi, Hanlin Xu, and Linfeng Zhang. 2025. Self speculative decoding for diffusion large language models. *arXiv preprint arXiv:2510.04147*.

Shansan Gong, Shivam Agarwal, Yizhe Zhang, Jiacheng Ye, Lin Zheng, Mukai Li, Chenxin An, Peilin Zhao, Wei BI, Jiawei Han, Hao Peng, and Lingpeng Kong. 2025. Scaling diffusion language models via adaptation from autoregressive models. In *International Conference on Learning Representations*, volume 2025, pages 5046–5073.

Shansan Gong, Mukai Li, Jiangtao Feng, Zhiyong Wu, and LingPeng Kong. 2023. Diffuseq: Sequence to sequence text generation with diffusion models. In *The Eleventh International Conference on Learning Representations*.

Daehoon Gwak, Minseo Jung, Junwoo Park, Minho Park, ChaeHun Park, Junha Hyung, and Jaegul Choo. 2025. Reward-weighted sampling: Enhancing non-autoregressive characteristics in masked diffusion LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 34574–34594, Suzhou, China. Association for Computational Linguistics.

Boyan Han, Yiwei Wang, Yi Song, Yujun Cai, and Chi Zhang. 2026. Dynamic infilling anchors for format-constrained generation in diffusion large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 26213–26227.

Xiaochuang Han, Sachin Kumar, and Yulia Tsvetkov. 2023. SSD-LM: Semi-autoregressive simplex-based diffusion language model for text generation and modular control. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11575–11596, Toronto, Canada. Association for Computational Linguistics.

Guangxin He, Shen Nie, Fengqi Zhu, Yuankang Zhao, Tianyi Bai, Ran Yan, Jie Fu, Chongxuan Li, and Binhang Yuan. 2026. UltraLLaDA: Scaling the context length to 128k for diffusion large language models. In *The Fourteenth International Conference on Learning Representations*.

Feng Hong, Geng Yu, Yushi Ye, Haicheng Huang, Huangjie Zheng, Ya Zhang, Yanfeng Wang, and Jiangchao Yao. 2026. Wide-in, narrow-out: Revokable decoding for efficient and effective DLLMs. In *The Fourteenth International Conference on Learning Representations*.

Lanxiang Hu, Zhaoxiang Feng, Yulun Wu, Haoran Yuan, Yujie Zhao, Yu-Yang Qian, Bojun Wang, Peng Zhao, Daxin Jiang, Yibo Zhu, et al. 2026a. Jetspec: Breaking the scaling ceiling of speculative decoding with parallel tree drafting. *arXiv preprint arXiv:2606.18394*.

Yuezhou Hu, Harman Singh, Monishwaran Maheswaran, Haocheng Xi, Coleman Hooper, Jintao Zhang, Aditya Tomar, Michael W Mahoney, Sewon Min, Mehrdad Farajtabar, et al. 2026b. Residual context diffusion language models. *arXiv preprint arXiv:2601.22954*.

Chihan Huang and Hao Tang. 2025. Ctrldiff: Boosting large diffusion language models with dynamic block prediction and controllable generation. *arXiv preprint arXiv:2505.14455*.

Jianuo Huang, Yaojie Zhang, Yicun Yang, Benhao Huang, Biqing Qi, Dongrui Liu, and Linfeng Zhang. 2025. Mask tokens as prophet: Fine-grained cache eviction for efficient dllm inference. *arXiv preprint arXiv:2510.09309*.

Jianuo Huang, Yaojie Zhang, Qituan Zhang, Hao Lin, Hanlin Xu, and Linfeng Zhang. 2026. Domino: Decoupling causal modeling from autoregressive drafting in speculative decoding. *arXiv preprint arXiv:2605.29707*.

Inclusion AI, Tiwei Bie, Haoxing Chen, Tieyuan Chen, Zhenglin Cheng, Long Cui, Kai Gan, Zhicheng Huang, Zhenzhong Lan, Haoquan Li, et al. 2026. Llada2. 0-uni: Unifying multimodal understanding and generation with diffusion large language model. *arXiv preprint arXiv:2604.20796*.

Daniel Israel, Guy Van den Broeck, and Aditya Grover. 2025. Accelerating diffusion llms via adaptive parallel decoding. In *Advances in Neural Information Processing Systems*, volume 38, pages 52870–52888. Curran Associates, Inc.

Metod Jazbec, Theo X Olausson, Louis Béthune, Pierre Ablin, Michael Kirchhof, João Monteiro, Victor Turrisi, Jason Ramapuram, and Marco Cuturi. 2025. Learning unmasking policies for diffusion language models. *arXiv preprint arXiv:2512.09106*.

Yuchu Jiang, Yue Cai, Xiangzhong Luo, Jiale Fu, Jiarui Wang, Chonghan Liu, and Xu Yang. 2026. d^{2}cache: Accelerating diffusion-based LLMs via dual adaptive caching. In *The Fourteenth International Conference on Learning Representations*.

Wonjun Kang, Kevin Galim, Seunghyuk Oh, Minjae Lee, Yuchen Zeng, Shuibai Zhang, Coleman Richard Charles Hooper, Yuezhou Hu, Hyung Il Koo, Nam Ik Cho, and Kangwook Lee. 2026. Parallelbench: Understanding the trade-offs of parallel decoding in diffusion LLMs. In *The Fourteenth International Conference on Learning Representations*.

BumJun Kim, Dongjae Jeon, Dueun Kim, Wonje Jeung, and Albert No. 2026. Rainbow padding: Mitigating early termination in instruction-tuned diffusion LLMs. In *The Fourteenth International Conference on Learning Representations*.

Jaeyeon Kim, Lee Cheuk-Kit, Carles Domingo-Enrich, Yilun Du, Sham Kakade, Timothy Ngotiaoco, Sitan Chen, and Michael Albergo. 2025. Any-order flexible length masked diffusion. *arXiv preprint arXiv:2509.01025*.

Fanheng Kong, Jingyuan Zhang, Yahui Liu, Zirui Wu, Yu Tian, Victoria W., and Guorui Zhou. 2025. Accelerating diffusion llm inference via local determinism propagation. *arXiv preprint arXiv:2510.07081*.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, SOSP '23, pages 611––626, New York, NY, USA. Association for Computing Machinery.

Duo Li, Zuhao Yang, Xiaoqin Zhang, Ling Shao, and Shijian Lu. 2025a. A comprehensive study on visual token redundancy for discrete diffusion-based multimodal large language models. *arXiv preprint arXiv:2511.15098*.

Guanghao Li, Zhihui Fu, Min Fang, Qibin Zhao, Ming Tang, Chun Yuan, and Jun Wang. 2025b. Diffuspec: Unlocking diffusion language models for speculative decoding. *arXiv preprint arXiv:2510.02358*.

Jinsong Li, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Jiaqi Wang, and Dahua Lin. 2026a. Beyond fixed: Training-free variable-length denoising for diffusion large language models. In *The Fourteenth International Conference on Learning Representations*.

Lijiang Li, Zuwei Long, Yunhang Shen, Heting Gao, Haoyu Cao, Xing Sun, Caifeng Shan, Ran He, and Chaoyou Fu. 2026b. Omni-diffusion: Unified multimodal understanding and generation with masked discrete diffusion. *arXiv preprint arXiv:2603.06577*.

Pengxiang Li, Yefan Zhou, Dilxat Muhtar, Lu Yin, Shilin Yan, Li Shen, Yi Liang, Soroush Vosoughi, and Shiwei Liu. 2026c. Diffusion language model knows the answer before it decodes. In *The Fourteenth International Conference on Learning Representations*.

Shufan Li, Jiuxiang Gu, Kangning Liu, Zhe Lin, Zijun Wei, Aditya Grover, and Jason Kuen. 2025c. Lavida-o: Elastic large masked diffusion models for unified multimodal understanding and generation. *arXiv preprint arXiv:2509.19244*.

Shufan Li, Jiuxiang Gu, Kangning Liu, Zhe Lin, Zijun Wei, Aditya Grover, and Jason Kuen. 2025d. Sparselavida: Sparse multimodal discrete diffusion language models. *arXiv preprint arXiv:2512.14008*.

Tianyi Li, Mingda Chen, Bowei Guo, and Zhiqiang Shen. 2025e. A survey on diffusion language models. *arXiv preprint arXiv:2508.10875*.

Yihao Liang, Ze Wang, Hao Chen, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Emad Barsoum, Zicheng Liu, and Niraj K Jha. 2026. Cd4lm: Consistency distillation and adaptive decoding for diffusion language models. *arXiv preprint arXiv:2601.02236*.

Haokun Lin, Xinle Jia, Haobo Xu, Bingchen Yao, Xianglong Guo, Yichen Wu, Zhichao Lu, Ying Wei, Qingfu Zhang, and Zhenan Sun. 2026. Duquant++: Fine-grained rotation enhances microscaling fp4 quantization. *arXiv preprint arXiv:2604.17789*.

Haokun Lin, Haobo Xu, Yichen Wu, Ziyu Guo, Renrui Zhang, Zhichao Lu, Ying Wei, Qingfu Zhang, and Zhenan Sun. 2025. Quantization meets dllms: A systematic study of post-training quantization for diffusion llms. *arXiv preprint arXiv:2508.14896*.

Aiwei Liu, Minghua He, Shaoxun Zeng, Sijun Zhang, Linhao Zhang, Chuhan Wu, Wei Jia, Yuan Liu, Xiao Zhou, and Jie Zhou. 2025a. Wedlm: Reconciling diffusion language models with standard causal attention for fast inference. *arXiv preprint arXiv:2512.22737*.

Fuliang Liu, Xue Li, Ketai Zhao, Yinxi Gao, Ziyan Zhou, Zhonghui Zhang, Zhibin Wang, Wanchun Dou, Sheng Zhong, and Chen Tian. 2026a. Dart: Diffusion-inspired speculative decoding for fast llm inference. *arXiv preprint arXiv:2601.19278*.

Xiaoran Liu, Yuerong Song, Zhigeng Liu, Zengfeng Huang, Qipeng Guo, Ziwei He, and Xipeng Qiu. 2026b. Longllada: Unlocking long context capabilities in diffusion llms. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 32186–32194.

Yangzhou Liu, Yue Cao, Hao Li, Gen Luo, Zhe Chen, Weiyun Wang, Xiaobo Liang, Biqing Qi, Lijun Wu, Changyao Tian, et al. 2025b. Sequential diffusion language models. *arXiv preprint arXiv:2509.24007*.

Yue Liu, Yuzhong Zhao, Zheyong Xie, Qixiang Ye, Jianbin Jiao, Yao Hu, Shaosheng Cao, and Yunfan Liu. 2026c. Balancing understanding and generation in discrete diffusion models. *arXiv preprint arXiv:2602.01362*.

Zhiyuan Liu, Yicun Yang, Yaojie Zhang, Junjie Chen, Chang Zou, Qingyan Wei, Shaobo Wang, Yichen Zhu, and Linfeng Zhang. 2025c. dllm-cache: Accelerating diffusion large language models with adaptive caching. *arXiv preprint arXiv:2506.06295*.

Guanxi Lu, Hao Mark Chen, Yuto Karashima, Zhican Wang, Daichi Fujiki, and Hongxiang Fan. 2026. Adablock-dLLM: Semantic-aware diffusion LLM inference via adaptive block size. In *The Fourteenth International Conference on Learning Representations*.

Linrui Ma, Yufei Cui, Kai Han, and Yunhe Wang. 2026a. Diffusion in diffusion: Reclaiming global coherence in semi-autoregressive diffusion. *arXiv preprint arXiv:2601.13599*.

Linrui Ma, Yufei Cui, Kai Han, and Yunhe Wang. 2026b. Mask is what dllm needs: A masked data training paradigm for diffusion llms. *arXiv preprint arXiv:2603.15803*.

Xinyin Ma, Runpeng Yu, Gongfan Fang, and Xinchao Wang. 2025a. dkv-cache: The cache for diffusion language models. In *Advances in Neural Information Processing Systems*, volume 38, pages 149009–149033. Curran Associates, Inc.

Yuxin Ma, Lun Du, Lanning Wei, Kun Chen, Qian Xu, Kangyu Wang, Guofeng Feng, Guoshan Lu, Lin Liu,

Xiaojing Qi, et al. 2025b. dinfer: An efficient inference framework for diffusion language models. *arXiv preprint arXiv:2510.08666*.

Amr Mohamed, Yang Zhang, Michalis Vazirgiannis, and Guokan Shang. 2025. Fast-decoding diffusion language models via progress-aware confidence schedules. *arXiv preprint arXiv:2512.02892*.

Amin Karimi Monsefi, Nikhil Bhendawade, Manuel Rafael Ciosici, Dominic Culver, Yizhe Zhang, and Irina Belousova. 2025. Fs-dfm: Fast and accurate long text generation with few-step diffusion language models. *arXiv preprint arXiv:2509.20624*.

John Nguyen, Marton Havasi, Tariq Berrada, Luke Zettlemoyer, and Ricky TQ Chen. 2025. Oneflow: Concurrent mixed-modal and interleaved generation with edit flows. *arXiv preprint arXiv:2510.03506*.

Quan Nguyen-Tri, Mukul Ranjan, and Zhiqiang Shen. 2026. Attention is all you need for KV cache in diffusion LLMs. In *The Fourteenth International Conference on Learning Representations*.

Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. 2025. Large language diffusion models. In *Advances in Neural Information Processing Systems*, volume 38, pages 50608–50646. Curran Associates, Inc.

Yidong Ouyang, Panwen Hu, Zhengyan Wan, Zhe Wang, Liyan Xie, Dmitriy Bespalov, Ying Nian Wu, Guang Cheng, Hongyuan Zha, and Qiang Sun. 2026. Training-free self-correction for multimodal masked diffusion models. *arXiv preprint arXiv:2602.02927*.

Han Peng, Peiyu Liu, Zican Dong, Daixuan Cheng, Junyi Li, Yiru Tang, Shuo Wang, and Wayne Xin Zhao. 2025. How efficient are diffusion language models? a critical examination of efficiency evaluation practices. *arXiv preprint arXiv:2510.18480*.

Yu-Yang Qian, Junda Su, Lanxiang Hu, Peiyuan Zhang, Zhijie Deng, Peng Zhao, and Hao Zhang. 2026a. d3llm: Ultra-fast diffusion llm using pseudo-trajectory distillation. *arXiv preprint arXiv:2601.07568*.

Yu-Yang Qian, Hao-Cong Wu, Chen Chen, Jiacheng Sun, Zhenhua Dong, Peng Zhao, and Zhi-Hua Zhou. 2026b. Adaflash: Adaptive speculative decoding via on-policy distilled diffusion drafters. *arXiv preprint arXiv:2607.19223*.

Subham Sekhar Sahoo, Jean-Marie Lemercier, Zhihan Yang, Justin Deschenaux, Jingyu Liu, John Thickstun, and Ante Jukic. 2026. Scaling beyond masked diffusion language models. *arXiv preprint arXiv:2602.15014*.

Jucheng Shen, Gaurav Sarkar, Yeonju Ro, Sharath Nittur Sridhar, Zhangyang Wang, Aditya Akella, and Souvik Kundu. 2025. Improving the throughput of diffusion-based large language models via a training-free confidence-aware calibration. *arXiv preprint arXiv:2512.07173*.

Yangyi Shen, Tianjian Feng, Jiaqi Han, Wen Wang, Tianlang Chen, Chunhua Shen, Jure Leskovec, and Stefano Ermon. 2026. Improving diffusion language model decoding through joint search in generation order and token space. *arXiv preprint arXiv:2601.20339*.

Qingyu Shi, Jinbin Bai, Zhuoran Zhao, Wenhao Chai, Kaidong Yu, Jianzong Wu, Yunhai Tong, Xiangtai Li, Xuelong Li, and Shuicheng Yan. 2025. Muddit: Liberating generation beyond text-to-image with a unified discrete diffusion model. *arXiv preprint arXiv:2505.23606*.

Yuerong Song, Xiaoran Liu, Ruixiao Li, Zhigeng Liu, Zengfeng Huang, Qipeng Guo, Ziwei He, and Xipeng Qiu. 2026. Sparse-dllm: Accelerating diffusion llms with dynamic cache eviction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 33038–33046.

Yuchuan Tian, Yuchen Liang, Shuo Zhang, Yingte Shu, Guangwen Yang, Wei He, Sibo Fang, Tianyu Guo, Kai Han, Chao Xu, et al. 2025. From next-token to next-block: A principled adaptation path for diffusion llms. *arXiv preprint arXiv:2512.06776*.

Caio Vicentino. 2026. Autoregressive vs. masked diffusion language models: A controlled comparison. *arXiv preprint arXiv:2603.22075*.

Xinyun Wang, Min Zhang, Sen Cui, Zhikang Chen, Bo Jiang, Kun Kuang, and Mingbao Lin. 2026. Reversible diffusion decoding for diffusion language models. *arXiv preprint arXiv:2602.00150*.

Linye Wei, Wenjue Chen, Pingzhi Tang, Xiaotian Guo, Le Ye, Runsheng Wang, and Meng Li. 2025. Orchestrating dual-boundaries: An arithmetic intensity inspired acceleration framework for diffusion language models. *arXiv preprint arXiv:2511.21759*.

Qingyan Wei, Yaojie Zhang, Zhiyuan Liu, Puyu Zeng, Yuxuan Wang, Biqing Qi, Dongrui Liu, and Linfeng Zhang. 2026. Accelerating diffusion large language models with slowfast sampling: The three golden principles. In *The Fourteenth International Conference on Learning Representations*.

Chengyue Wu, Hao Zhang, Shuchen Xue, Shizhe Diao, Yonggan Fu, Zhijian Liu, Pavlo Molchanov, Ping Luo, Song Han, and Enze Xie. 2026a. Fast-dLLM v2: Efficient block-diffusion LLM. In *The Fourteenth International Conference on Learning Representations*.

Chengyue Wu, Hao Zhang, Shuchen Xue, Zhijian Liu, Shizhe Diao, Ligeng Zhu, Ping Luo, Song Han, and Enze Xie. 2026b. Fast-dLLM: Training-free acceleration of diffusion LLM by enabling KV cache and parallel decoding. In *The Fourteenth International Conference on Learning Representations*.

Shutong Wu and Jiawei Zhang. 2025. Free draft-and-verification: Toward lossless parallel decoding for diffusion large language models. *arXiv preprint arXiv:2510.00294*.

Tong Wu, Zhihao Fan, Xiao Liu, Hai-Tao Zheng, Yeyun Gong, yelong shen, Jian Jiao, Juntao Li, zhongyu wei, Jian Guo, Nan Duan, and Weizhu Chen. 2023. Ar-diffusion: Auto-regressive diffusion model for text generation. In *Advances in Neural Information Processing Systems*, volume 36, pages 39957–39974. Curran Associates, Inc.

Yi Xin, Qi Qin, Siqi Luo, Kaiwen Zhu, Juncheng Yan, Yan Tai, Jiayi Lei, Yuewen Cao, Keqi Wang, Yibin Wang, et al. 2025. Lumina-dimoo: An omni diffusion large language model for multi-modal generation and understanding. *arXiv preprint arXiv:2510.06308*.

Chen Xu and Dawei Yang. 2025. Dllmquant: Quantizing diffusion-based large language models. *arXiv preprint arXiv:2508.14090*.

Chenkai Xu, Yijie Jin, Jiajun Li, Yi Tu, Guoping Long, Dandan Tu, Mingcong Song, Hongjie Si, Tianqi Hou, Junchi Yan, et al. 2025. Lopa: Scaling dllm inference via lookahead parallel decoding. *arXiv preprint arXiv:2512.16229*.

Haobo Xu, Sirui Chen, Yuanchen Bei, Lingjie Chen, Yuchen Yan, Dongqi Fu, Jingrui He, and Hanghang Tong. 2026. Predict, don't iterate: Efficient adaptive-length infilling for diffusion language models. *arXiv preprint arXiv:2609.02108*.

Kaisen Yang, Jayden Teoh, Kaicheng Yang, Yitong Zhang, and Alex Lamb. 2026a. Improving sampling for masked diffusion models via information gain. *arXiv preprint arXiv:2602.18176*.

Ling Yang, Ye Tian, Bowen Li, Xinchen Zhang, Ke Shen, Yunhai Tong, and Mengdi Wang. 2025a. Mmada: Multimodal large diffusion language models. In *Advances in Neural Information Processing Systems*, volume 38, pages 138867–138907. Curran Associates, Inc.

Wanqi Yang, Yuexiao Ma, Alexander Conzelmann, Xiawu Zheng, Michael W Mahoney, T Konstantin Rusch, and Shiwei Liu. 2026b. Alphaq: Calibration-free bit allocation for mixture-of-experts quantization. *arXiv preprint arXiv:2606.04980*.

Yicun Yang, Cong Wang, Shaobo Wang, Zichen Wen, Biqing Qi, Hanlin Xu, and Linfeng Zhang. 2025b. Diffusion llm with native variable generation lengths: Let [eos] lead the way. *arXiv preprint arXiv:2510.24605*.

Bingchen Yao, Haobo Xu, Haokun Lin, Yichen Wu, Ziyu Guo, Renrui Zhang, Zhichao Lu, Zhenan Sun, and Ying Wei. 2026. Stepquant: When and where errors matter in delta-rule recurrent state quantization. *arXiv preprint arXiv:2609.38169*.

Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui Wu, Xin Jiang, Zhenguo Li, and Lingpeng Kong. 2025. Dream 7b: Diffusion large language models. *arXiv preprint arXiv:2508.15487*.

Yushi Ye, Feng Hong, Huangjie Zheng, Xu Chen, Zhiyong Chen, Yanfeng Wang, and Jiangchao Yao. 2026. Rejection mixing: Fast semantic propagation of mask tokens for efficient dllm inference. *arXiv preprint arXiv:2602.22868*.

Zebin You, Shen Nie, Xiaolu Zhang, Jun Hu, Jun Zhou, Zhiwu Lu, Ji-Rong Wen, and Chongxuan Li. 2025. Llada-v: Large language diffusion models with visual instruction tuning. *arXiv preprint arXiv:2505.16933*.

Zebin You, Xiaolu Zhang, Jun Zhou, Chongxuan Li, and Ji-Rong Wen. 2026. Llada-o: An effective and length-adaptive omni diffusion model. *arXiv preprint arXiv:2603.01068*.

Runpeng Yu, Qi Li, and Xinchao Wang. 2025a. Discrete diffusion in large language and multimodal models: A survey. *arXiv preprint arXiv:2506.13759*.

Runpeng Yu, Xinyin Ma, and Xinchao Wang. 2025b. Dimple: Discrete diffusion multimodal large language model with parallel decoding. *arXiv preprint arXiv:2505.16990*.

Tianao Zhang, Zhiteng Li, Xianglong Yan, Haotong Qin, Yong Guo, and Yulun Zhang. 2026. Quant-dLLM: Post-training extreme low-bit quantization for diffusion large language models. In *The Fourteenth International Conference on Learning Representations*.

Jianyuan Zhong, Kaibo Wang, Ding Ding, Zijin Feng, Haoli Bai, Yang Xiang, Jiacheng Sun, and Qiang Xu. 2026. Stabilizing reinforcement learning for diffusion language models. *arXiv preprint arXiv:2603.06743*.

Xueyu Zhou, Yangrong Hu, and Jian Huang. 2026a. Dos: Dependency-oriented sampler for masked diffusion language models. *arXiv preprint arXiv:2603.15340*.

Zhanhui Zhou, Lingjie Chen, Hanghang Tong, and Dawn Song. 2026b. dllm: Simple diffusion language modeling. *arXiv preprint arXiv:2602.22661*.

Fengqi Zhu, Rongzhen Wang, Shen Nie, Xiaolu Zhang, Chunwei Wu, Jun Hu, Jun Zhou, Jianfei Chen, Yankai Lin, Ji-Rong Wen, et al. 2025a. Llada 1.5: Variance-reduced preference optimization for large language diffusion models. *arXiv preprint arXiv:2505.19223*.

Fengqi Zhu, Zebin You, Yipeng Xing, Zenan Huang, Lin Liu, Yihong Zhuang, Guoshan Lu, Kangyu Wang, Xudong Wang, Lanning Wei, et al. 2025b. Llada-moe: A sparse moe diffusion language model. *arXiv preprint arXiv:2509.24389*.

Ying Zhu, Jiaxin Wan, Xiaoran Liu, Siyang He, Qiqi Wang, Xu Guo, Tianyi Liang, Zengfeng Huang, Ziwei He, and Xipeng Qiu. 2025c. Dirl: An efficient post-training framework for diffusion language models. *arXiv preprint arXiv:2512.22234*.

A Multimodal DLM Efficiency

Training Stage. To alleviate the inherent token-by-token decoding latency of autoregressive multimodal models, recent work has explored multimodal diffusion language models (MM-dLLMs) as a parallel and more controllable generation paradigm (Yang et al., 2025a; You et al., 2025; Shi et al., 2025; Li et al., 2025c; You et al., 2026; Li et al., 2026b; Inclusion AI et al., 2026). Dimple (Yu et al., 2025b) develops an MM-dLLM by employing a hybrid "autoregressive-then-diffusion" training paradigm to stabilize convergence. Pushing block-wise efficiency, SDAR-VL (Cheng et al., 2025b) introduces an asynchronous noise scheduling strategy, demonstrating that block-wise discrete diffusion can outperform strong autoregressive baselines in vision-language understanding tasks.

Inference Stage. To address the computational intensity of multimodal denoising, recent works propose acceleration from algorithmic perspectives. D^{3}ToM (Chang et al., 2025) dynamically merges redundant visual tokens via "decider-guided" importance maps, while Sparse-LaViDa (Li et al., 2025d) truncates unnecessary masked tokens by utilizing specialized register tokens for compact representation. Complementing these, Li et al. (2025a) empirically analyze pruning-induced information loss and show that from-scratch dMLLMs can recover part of this loss on long-answer tasks, motivating progressive or late-step pruning under suitable conditions. Furthermore, Dimple optimizes the inference pipeline through "confident decoding," which dynamically adjusts the number of tokens generated per step to reduce iterations by two-thirds, and re-incorporates the prefilling mechanism from AR models to jointly minimize generation complexity. Lumina-DiMOO (Xin et al., 2025) applies dLLM efficiency to multi-modal generation using a "Max-Logit" cache, freezing representations of tokens with high prediction confidence to accelerate image-text synthesis. FreeCorrection (Ouyang et al., 2026) adds a training-free self-correction loop on top of multimodal masked diffusion, rolling back wrongly committed tokens guided by the model's own confidence.

Discussion. Together, these advancements transition multimodal dLLMs from theoretical exploration to high-performance, sparse-inference systems. However, the rapid development of MM-dLLMs still necessitates multi-level optimizations, spanning from data-centric noise scheduling to hardware-aware kernel designs, to fully unlock their potential in real-time, large-scale multimodal applications. This remains a fertile ground for future research into unified, cross-modal diffusion architectures.

B Quantitative Comparison of Inference Acceleration Methods

Table 1 consolidates reported performance from representative inference acceleration methods discussed in Section 3. We mainly select methods that have been accepted at top venues (ICLR, NeurIPS, ICML, AAAI) or have accumulated significant community attention. All numbers are taken directly from original papers; differences in evaluation protocols (prompt length, sampling temperature, number of denoising steps) mean that cross-method comparisons should be interpreted as indicative rather than strictly controlled.

Because the entries are not measured under a shared protocol, we read Table 1 for direction rather than for magnitude. Four trends are consistent enough to survive this caveat. First, learning-based parallel decoding methods (dParallel, Learn2PD, d3LLM) report the highest throughput gains, but each requires additional training or distillation, so the gain is not free and the reported factors depend heavily on the chosen operating point. Second, training-free cache methods (dKV-Cache, d^{2}Cache, Sparse-dLLM) report more modest gains that appear robust across base models, and they compose with parallel decoding, as suggested by dLLM-Cache when the two are combined. Third, speculative verification (Spiffy, DEER) is the only family that reports lossless output, which follows from its draft-then-verify construction rather than from empirical tuning; this makes it attractive when quality cannot be traded away. Finally, extreme quantization (Quant-dLLM at 2-bit) trades moderate accuracy for aggressive memory reduction, opening doors for on-device dLLM inference. We stress that these are directional observations, not a ranking: establishing which family is fastest under matched conditions would require a unified re-evaluation, which we discuss in the Limitations.

C Comparison with Existing Surveys

Table 2 compares our survey with two representative dLLM surveys. Yu et al. (2025a) covers

Method	Category	TF	Base Model	Speedup	ΔAcc	Venue
Parallel Decoding — Heuristic						
Fast-dLLM (Wu et al., 2026b)	Confidence + Cache	✓	LLaDA-8B	7.6×†	≈0	ICLR'26
SlowFast (Wei et al., 2026)	Property-Informed	✓	LLaDA-8B	3.2–15.6×	≈0	ICLR'26
WINO (Hong et al., 2026)	Draft-and-Verify	✓	LLaDA-8B	2.5–3.8×	+1–3%	ICLR'26
Parallel Decoding — Learning-based						
dParallel (Chen et al., 2026b)	Certainty Distill.	✗	LLaDA-8B	8.5–10.5×	≈0	ICLR'26
Learn2PD (Bao et al., 2026)	Learnable Filter	✗	LLaDA-8B	22.6×	≈0	ICLR'26
d3LLM (Qian et al., 2026a)	Pseudo-Traj. Distill.	✗	LLaDA-8B / Dream-7B	5× vs. AR	≈0	ICML'26
Cache Management						
dKV-Cache (Ma et al., 2025a)	Delayed Caching	✓	LLaDA-8B / Dream-7B	2–10×	≈0	NeurIPS'25
d^2Cache (Jiang et al., 2026)	Dual Adaptive	✓	LLaDA-8B / Dream-7B	1.9–4.8×	≈0	ICLR'26
dLLM-Cache (Liu et al., 2025c)	Similarity Verify	✓	LLaDA-8B / Dream-7B	9.1×(FLOPs ↓)†	≈0	ICML'26
Sparse-dLLM (Song et al., 2026)	Dynamic Eviction	✓	LLaDA-8B / Dream-7B	≤10×	≈0	AAAI'26
Compression & Speculative Decoding						
Spiffy (Agrawal et al., 2025)	Self-Speculation	✓	LLaDA-8B	2.8–7.9×†	Lossless	–
DEER (Cheng et al., 2025c)	dLLM-AR Synergy	✗	Qwen3-30B (target)	5.5×	Lossless	–
Quant-dLLM (Zhang et al., 2026)	W2 Quantization	✓	LLaDA-8B / Dream-7B	4.4× mem.	−7–11%	ICLR'26

Table 1: Reported inference acceleration results for representative dLLM methods. "TF" = training-free. Speedup is wall-clock throughput ratio vs. vanilla dLLM decoding from corresponding papers. "ΔAcc" denotes accuracy change relative to the unaccelerated baseline on the same model; "≈0" means within noise (≤0.5%). "–" indicates not reported. †Combined with parallel decoding. ‡Combined with dLLM-Cache.

	Yu et al. (2025a)	Li et al. (2025e)	**Ours**
Focus	General	General	**Efficiency**
Time window	≤25/09	≤25/12	**≤26/09**
Efficiency taxonomy	✗	△	✓
Training (AR→dLLM)	△	△	✓
Decoding & cache	△	✓	✓
Compression & SD	△	△	✓
Context scalability	✗	△	✓
Frameworks & serving	✗	△	✓
Multimodal efficiency	△	△	✓

Table 2: Comparison with existing dLLM surveys. ✓ = dedicated, △ = partial, ✗ = not covered.

the foundations, trustworthiness, and applications of discrete diffusion across language and multimodal models, but do not provide a dedicated efficiency taxonomy or systematically discuss cache management, serving frameworks, or long-context methods. Li et al. (2025e) offers a broader DLM survey that touches on caching and decoding parallelism, yet treats efficiency as one of many topics rather than the central axis; AR-to-dLLM conversion, compression, variable-length generation, and serving receive limited coverage. Our work is *efficiency-first*: we organize the literature along the dLLM lifecycle (Training, Inference, Context, Framework, Multimodality), with each category targeting a concrete deployment bottleneck.

D Cross-Category Interactions

Our taxonomy separates the lifecycle stages so that each can be surveyed against its own bottleneck. The separation also reads in reverse: when two categories act on the same quantity, tuning them independently can cancel the benefit of both. Table 3 lists the three couplings we find most consequential and are under-explored.

E Potential Risks and Scientific Artifacts

This survey does not introduce a new model, dataset, or executable system. Nevertheless, by consolidating techniques for improving the training, inference, and deployment efficiency of diffusion language models, it may indirectly lower

Interaction	Why they couple	Bottleneck	Open question
Compression × **Inference**	Decoders commit tokens by confidence; quantization perturbs the **logits** behind that score	Error rate rises **silently**: the trade-off measured at full precision no longer holds	Should the unmasking threshold scale with **bit-width**, or confidence be recalibrated post-quantization?
Inference × **Framework**	Samplers are trained under one **schedule**; serving varies block size and latency budget	Reported speedups are **schedule-specific** and degrade once deployment shifts	Should the sampler be **conditioned** on block size and latency budget, not a fixed schedule?
Context × **Cache**	Length control sets how much canvas holds **content**; cache policy sets what is kept	Uniform refresh and eviction pay for **padded** positions	Should refresh and eviction be **length-aware**, keyed on predicted answer length?

Table 3: Couplings between taxonomy categories, each acting on a shared quantity that no surveyed method optimizes jointly. Rows correspond to §3.3×§3.1, §3.1×§5, and §4.1×§3.2. These questions are stated as open problems.

the computational cost of deploying such models and thereby facilitate both beneficial applications and potential misuse at larger scale. Such risks are primarily inherited from the underlying models and their deployment settings. We also maintain an accompanying GitHub repository that serves as a research-oriented curated resource for efficient diffusion language models, including organized references and links to publicly available implementations when provided by the original authors. The repository is released under the CC BY-NC-ND 4.0 license and is intended to support literature navigation, reproducibility, and future research.

F Use of AI Assistants

The authors used AI-based writing assistants only for language polishing, grammar checking, and improving the clarity of presentation. All technical content, literature selection, analysis, and final writing decisions were reviewed and verified by the authors.